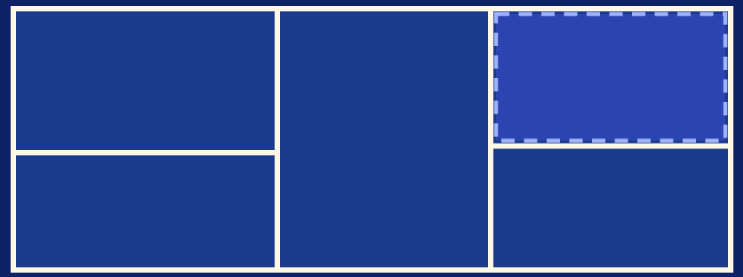
FUSEPLAN

Separating Complementarity from Ensemble Effects in Partial Floor Plan Retrieval

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PROBLEM

Floor-plan retrieval methods train and evaluate on complete plans. We study retrieval from **incomplete queries**.

Goal → from a partial plan, find its source among 67,405 RPLAN plans.

Question *→ does the image add information to the graph, or does it only act as a second model?*

RELATED WORK

LayoutGMN 2021 → graph matching on complete plans

LayoutGKN 2025 → **our baseline**

Compares the room graphs of two plans and learns how similar they are. Built for complete plans: we retrain it on partial ones.

SSIG 2023 → similarity of complete plans

Graph2Plan 2020 → partial plans, for generation only

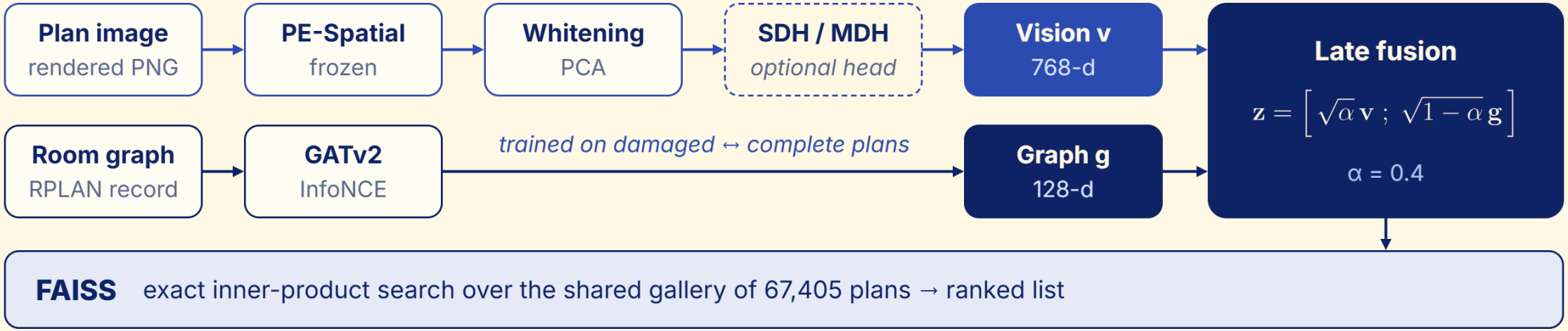
Sketch retrieval 2020 → **our metric**

Finds a photo from a sketch while it is still being drawn, scoring the search at each stage.

CrossOver, MMFE → end-to-end fusion, fused accuracy only

Gap: no retrieval method takes partial plans as queries.

METHOD



Graph branch

GATv2 trained to map a partial plan onto its complete version. Checkpoint chosen on robustness.

Vision branch

Frozen PE-Spatial + PCA whitening, optionally with a head trained on the damage:

SDH: room removal
MDH: rooms, crop and patch

Fusion score

$$\langle \mathbf{z}_q, \mathbf{z}_p \rangle = \alpha \langle \mathbf{v}_q, \mathbf{v}_p \rangle + (1-\alpha) \langle \mathbf{g}_q, \mathbf{g}_p \rangle$$

A weighted average of the two cosine similarities, α chosen on validation.

EVALUATION PROTOCOL

Self-recovery metric

Query: a plan with 25, 50 or 75% of its rooms removed.

$$\mathrm{RR}(q, f) = \begin{cases} 1/r_q(f) & \text{if } r_q(f) \leq 100 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathrm{AUC}(q) = \frac{1}{3} \sum_{f \in \{0.25, 0.5, 0.75\}} \mathrm{RR}(q, f)$$

$$\mathrm{AUC} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{AUC}(q)$$

AUC = 1 means the source always ranks first

Selection on validation robustness: not on the loss

Graph	configuration and checkpoint with the best room-removal AUC
Vision	encoder with the best AUC averaged over rooms, crop and patch
Fusion weight α	best room-removal AUC on validation

⚠ **Why?**

A lower loss does not mean better retrieval. So we select on the task itself, finding the source plan.

RESULTS

Gallery

67,405 RPLAN plans

Queries

test plans with 25, 50 or 75% of their rooms removed

Baselines

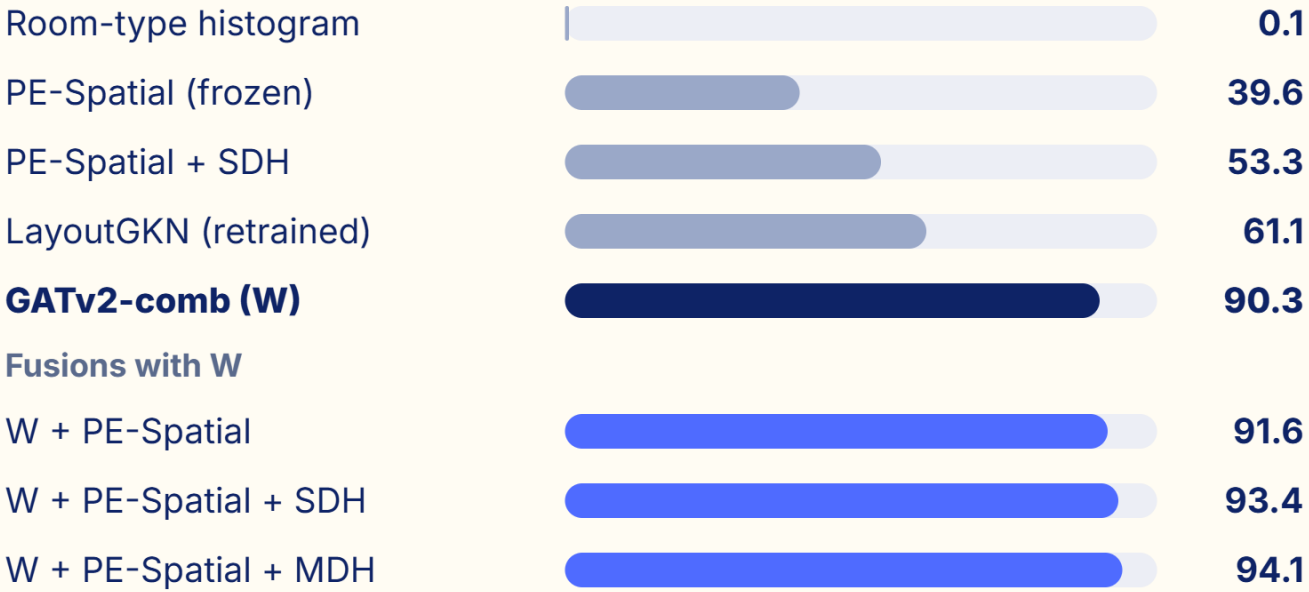
room-type histogram; LayoutGKN retrained on our task

Comparison

same queries for every system

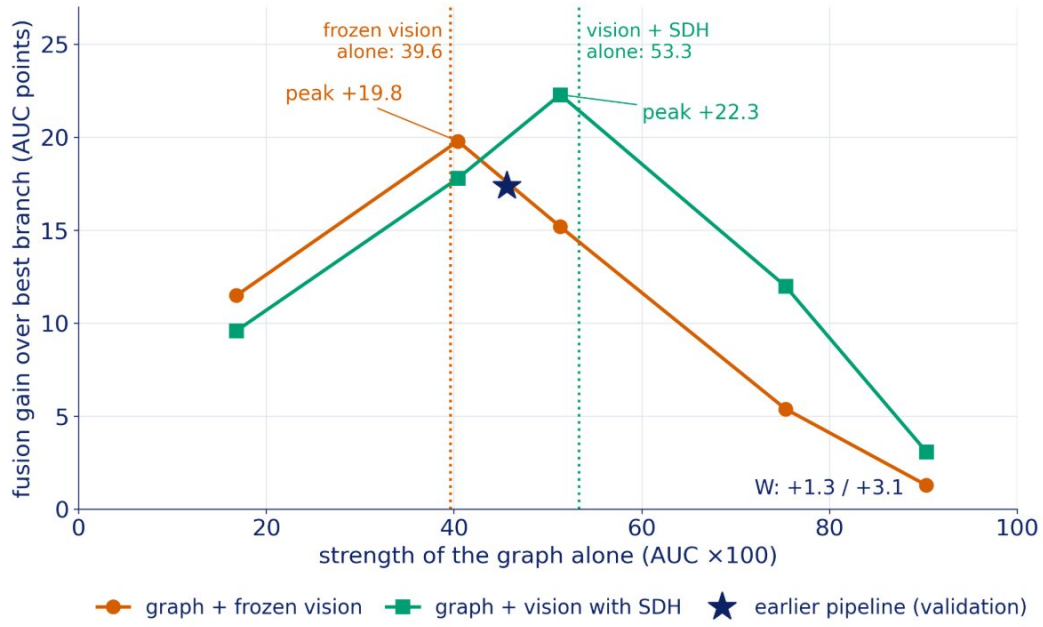
Robustness to room removal

test split, AUC ×100



W beats LayoutGKN by 29 points. Every fusion improves on W, mostly at 75% removal

RELATIVE STRENGTH & ABLATION



The fusion gain peaks when the two branches are equally strong.

Graph ablation (GATv2)

Reference	51.6
τ = 0.1 only	74.5
W (comb.)	**90.7**

Combining changes beats any single one

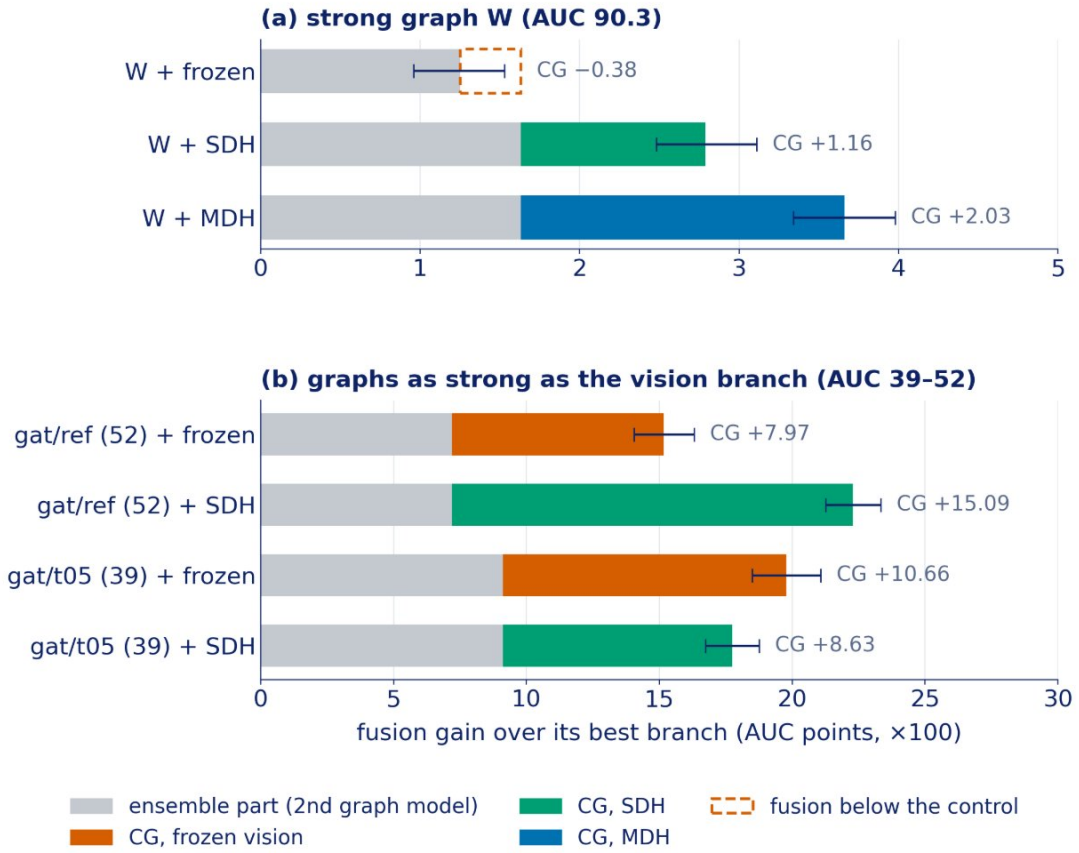
Vision ablation: raw → whitened

PE-Spatial	29.3 → 52.4
DINOv2	17.3 → 38.1
TIPSv2	11.1 → 31.7

Whitening helps every encoder

Graph: GCN, GraphSAGE, **GATv2**
Vision: 8 frozen encoders, **PE-Spatial** best

COMPLEMENTARITY GAIN



1 **Problem → ensemble effect**

Fusing any two models raises AUC: a gain alone does not prove the image adds information.

2 **Control tests**

Fuse W with a second graph model that holds the same information.

3 **CG = fusion gain – control gain**

$$\mathrm{gain}_F(q) = \mathrm{AUC}_F(q) - \mathrm{AUC}_b(q)$$

$$\mathrm{CG} = \mathrm{gain}_{V+G} - \mathrm{gain}_{C^\star}$$

b: better branch, C⋆: stronger control

Frozen vision: only an ensemble effect.
SDH / MDH: information of their own, yet the control explains about half of the gain.

CONCLUSIONS

Graph branch

Trained on damaged ↔ complete pairs, selected on robustness: AUC 90.3, +29 over LayoutGKN

RECOMMENDATION

Report every fusion gain next to an ensemble control

Vision branch

Frozen PE-Spatial 39.6; damage-trained heads: SDH 53.3, MDH 65.3

FUSION

Increases performance: 90.3 → 91.6 / 93.4 / 94.1

Ensemble effect: a 2nd graph model already gives +1.6

Complementarity gain: only with damage-trained heads, or a graph no stronger than the image

Limitations

- RPLAN only, synthetic damages
- graphs from exact metadata, not parsed images
- one global α; per-axis relevance is circular

Future improvements

- multiple, external datasets
- graphs parsed from images
- learned or per-query fusion; human relevance judgements

Thank you for your attention!

